



Government of Tamilnadu

Department of Employment and Training

Course : TNPSC Group II Exam
Subject : Aptitude and Mental Ability
Topic : **Area and Volume**

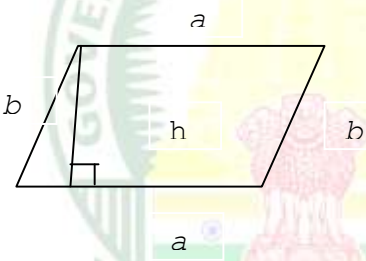
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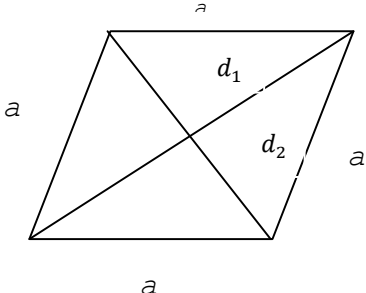
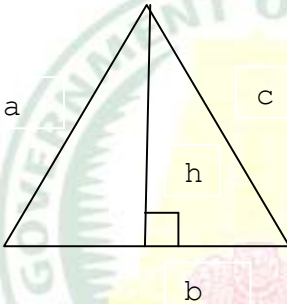
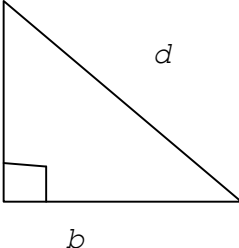
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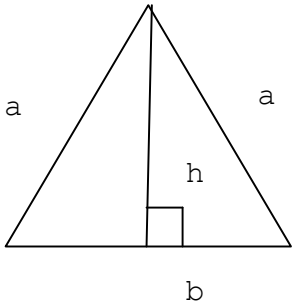
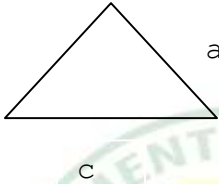
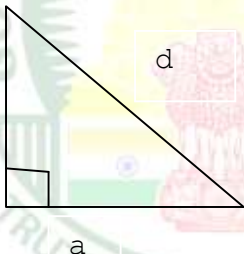
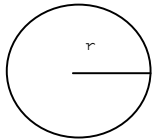
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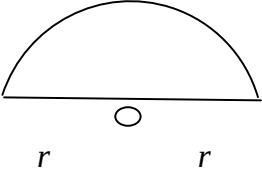
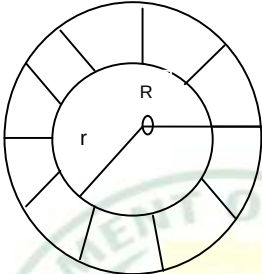
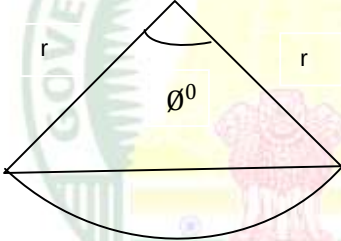
**Commissioner,
Department of Employment and Training.**

AREA AND VOLUME

S.No	Name	Figure	Perimeter in units of length	Area in square units
1	Rectangle	 a b a=length, b=breath.	$2(a + b)$	ab
2	Square	 a a = side	$4a$	a^2 $\frac{1}{2}(\text{diagonal})^2$
3	Parallelogram	 a b h a a=side, b=side adjacent, h = distance between the opp.parallel sides	$2(a + b)$	ah

4	Rhombus	 A= side of rhombus; d_1, d_2 Are the two diagonals.	4a	$\frac{1}{2} d_1 d_2$
5	Triangle	 b is the base and h is the altitude. a, b, c are three sides of	$a + b + c = 2s$ Where s is the semi perimeter	$\frac{1}{2} bh$ or $\sqrt{s(s-a)(s-b)(s-c)}$
6	Right triangle	 $d(\text{hypotenuse}) = \sqrt{b^2 + h^2}$	$b + h + d$	$\frac{1}{2} bh$

7	Equilateral Triangle		3a	i) $\frac{1}{2} ah$ ii) $\frac{\sqrt{3}}{4} a^2$
8	Isosceles triangle	 c = unequal side a = equal side	2a + c	$c \frac{\sqrt{4a^2 - c^2}}{4}$
9	Isosceles right triangle	 d(hypotenuse) = $a\sqrt{2}$ a = Each of equal sides. The angles are $90^\circ, 45^\circ, 45^\circ$	2a + d	$\frac{1}{2} a^2$
10	Circle	 r = radius of the circle $\pi = \frac{22}{7}$ or 3.1416	2 π r	πr^2

11	SemiCircle	 r r r = radius of the circle	$\pi r + 2r$	$\frac{1}{2} \pi r^2$
12	Ring(Shaded region)			$\pi(R^2 - r^2)$
13	Sector of a circle	 θ° = central angle of the sector R = radius of the sector l = length of the Arc	$l + 2r$ where $l = \frac{\theta}{360} \times 2\pi r$	$\frac{\theta}{360} \times \pi r^2$

Some theory questions related to geometry were also asked in TNPSC.

1. The angles opposite to two equal sides of a triangle are **EQUAL**.
2. If the angles of a triangle are in the ratio 1 : 2 : 2, then the triangle is **ISOSCELES**.
3. The polygon whose sum of interior angles equals 360° is **QUADRILATERAL**.

EXAMPLE PROBLEMS:

1. A circular ground of radius 7 has a path of width 7 m around it on its outside. The area of the path is approximately equal to _____ ($\pi = 22/7$)

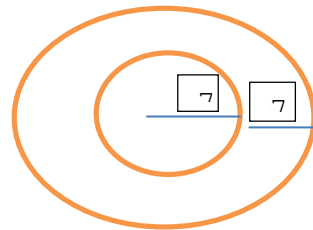
SOLUTION:

$$\begin{aligned}\text{Area of inner circle} &= \pi r^2 \\ &= \frac{22}{7} \times 7 \times 7 \quad (r = 7) \\ &= 154\end{aligned}$$

$$\begin{aligned}\text{Area of outer circle} &= \pi r^2 \\ &= \frac{22}{7} \times 14 \times 14 \quad (r = 7 + 7 = 14) \\ &= 616\end{aligned}$$

$$\begin{aligned}\text{Area of Path} &= 616 - 154 \\ &= 462\end{aligned}$$

$$\text{Ans} = 462 \text{ sq.m}$$



2. A cylindrical tank of diameter 28 cms is full of water. If 11 litres of water is drawn off, the water level in the tank will drop by

SOLUTION:

$$\text{Diameter} = 28 \text{ cms} \quad \text{Radius} = 14 \text{ cms}$$

$$\text{Volume of Cylinder} = \pi r^2 h$$

$$11 \text{ litres of water} = 11000 \text{ cms.}$$

$$\frac{22}{7} \times 14 \times 14 \times h = 1100$$

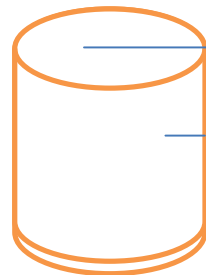
$$\pi r^2 h = 11000$$

$$\frac{22}{7} \times 14 \times 14 \times h = 11000$$

$$h = 11000 \times \frac{7}{22} \times \frac{1}{14} \times \frac{1}{7}$$

$$= \frac{125}{7}$$

$$\text{Ans:} = 17 \frac{6}{7} \text{ cm}$$



3. The length, breadth and height of a room are respectively 12 metres, 9 metres and 6 metres. How many cubic boxes are needed to fill the room if the side of each box is 1.5 metres?

SOLUTION:

$$\begin{aligned} L &= 12\text{m}, & B &= 9\text{m}, & H &= 6\text{m} \\ \text{Volume of Cuboid} &= l \times b \times h \text{ Cubic units} \\ &= 12 \times 9 \times 6 \\ &= 648 \text{ m}^3 \end{aligned}$$

$$\text{Volume of cube} = a^3$$

$$\text{Here } a = 1.5$$

$$\text{Volume} = (1.5)^3 = 3.375$$

$$\begin{aligned} \text{No. of cubic boxes} &= \frac{\text{Volume of cuboid}}{\text{Volume of cube}} \\ &= \frac{648}{3.375} \end{aligned}$$

$$\text{No. of cubic boxes} = 192$$

4. If the radius of a circle is increased by 25% then its area is increased by

SOLUTION:

$$\text{Let } R = 100$$

$$R \text{ Increased by } 25\%$$

$$\begin{aligned} \text{Area} &= \pi r^2 \\ &= \pi \times 100 \times 100 \\ &= 10000 \pi \\ &= \pi \times 125 \times 125 \end{aligned}$$

$$\frac{25}{100} \times 100 = 25$$

$$R_1 = 100 + 25 = 125$$

$$\text{Area} = \pi R_1^2$$

$$\text{Area} = 15625 \pi$$

$$\text{Increased \% area} = \frac{15625 \pi - 10000 \pi}{10000} \times 100$$

$$= \frac{5625}{100} \pi$$

$$= 56.25\% \text{ of } \pi$$

$$\text{Increased \%} = 56.25 \%$$

5. If the area of an equilateral triangle is $4\sqrt{3} \text{ m}^2$, then its perimeter is

SOLUTION:

$$\text{Area of Equilateral triangle} = \frac{\sqrt{3}}{4} a^2 = 4\sqrt{3}$$

$$\frac{\sqrt{3}}{4} a^2 = 4\sqrt{3}$$

$$a^2 = \frac{4\sqrt{3} \times 4}{\sqrt{3}}$$

$$a^2 = 4 \times 4 = 4^2$$

$$a = 4$$

$$\text{Perimeter of equilateral triangle} = 3a$$

$$= 3 \times 4$$

$$= 12 \text{ m}$$

6. The radius of sphere B is one fourth of radius of sphere A. Then the ratio of volume between A and B is.

SOLUTION:

$$\text{Radius A} = r$$

$$\text{Radius B} = \frac{1}{4}r$$

$$\text{Volume of Sphere} = \frac{4}{3} \pi r^3$$

$$\text{Volume A : Volume B}$$

$$\frac{4}{3} \pi r^3 : \frac{4}{3} \pi \left(\frac{1}{4}r\right)^3$$

$$r^3 : \frac{1}{64} r^3$$

$$1 : \frac{1}{64}$$

$$64 : 1 = \text{Ratio}$$

7. If the volume and surface area of a sphere are numerically the same, then its radius is

SOLUTION:

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$\text{Surface area of sphere} = 4 \pi r^2$$

Volume = Surface area

$$\frac{4}{3} \pi r^3 = 4 \pi r^2$$

$$\frac{1}{3} r^3 = r^2$$

$$\frac{r^3}{r^2} = 3$$

$$r = 3 \text{ units}$$

8. Area of a rectangle one of whose sides is 6 m and diagonal 10 m is

SOLUTION:

$$AC^2 = AB^2 + BC^2$$

$$10^2 = 6^2 + BC^2$$

$$100 = 36 + BC^2$$

$$100 - 36 = BC^2$$

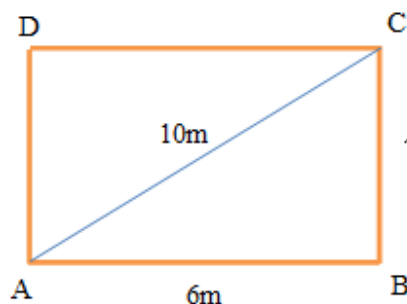
$$64 = BC^2$$

$$BC = 8$$

Area of rectangle = AB x BC

$$= 6 \times 8$$

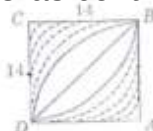
$$= 48 \text{ m}$$



EXERCISE PROBLEM:

1. In the adjoining figure of the arcs have A and C as centres. Find the area of shaded portions.

(A) 82 (B) 72 (C) 42 (D) 84

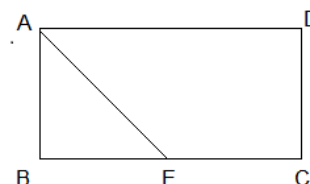


2. If the radius of a circle is increased by 25% then its area is increased by

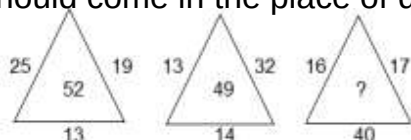
(A) 50% (B) 25% (C) 56.25% (D) 46.25%

3. In the adjoining figure ABCD is a rectangle and area of AABE is 15 cm^2 . If $EC = 2BE$, then area of rectangle is :

(A) 24 cm^2 (B) 48 cm^2 (C) 90 cm^2 (D) 120 cm^2



4. What should come in the place of question mark (?)



- (A) 63 (B) 73 (C) 53 (D) 83

5. When represented through a pie diagram, the angle that corresponds to rent is :

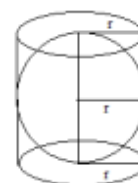
- (A) 32° (B) 48° (C) 72° (D) 96°

6. If the radius of a circle is decreased by 50%, find the % decrease in its area.

- (A) 74% (B) 75% (C) 76% (D) 95%

7. A sphere is placed inside a right circular hollow cylinder so as to touch the top, base and the lateral surface of the cylinder as shown in figure. If the radius of the sphere is r , the volume of the cylinder is

- (A) $4\pi r^3$ (B) $\frac{8\pi}{3}r^3$ (C) $2\pi r^3$ (D) $8\pi r^3$

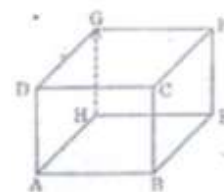


8. 12 spheres of the same size are made from melting a solid cylinder of 16 cm diameter and 2 cm height then the diameter of each sphere is

- (A) $\sqrt{3}$ cm (B) 2 cm (C) 3 cm (D) 4 cm

9. If the volume of the cube ABCDEFGH is 64 cu.cm. What is the shortest distance from D to E?

- (A) $4\sqrt{2}$ (B) $4\sqrt{3}$ (C) $4\sqrt{6}$ (D) $8\sqrt{2}$



10. The sum of the interior angles of a Hexagon is

- (A) 360° (B) 240° (C) 720° (D) 180°

11. Find the volume of a solid cylinder whose radius is 14 cm and height 30 cm

- (A) 18380 cm^3 (B) 18480 cm^3 (C) 18580 cm^3 (D) 18680 cm^3

12. If

□ → Mathematics

○ → Tamil

△ → English



Then the number of students studied Maths and Tamil alone.

- (A) 5 (B) 7 (C) 8 (D) 12

13. A rectangular carpet has an area of 60 sq.m. Its diagonal and longer side together equal 5 times the shorter side. The length of the carpet is

- (A) 5 m (B) 12 m (C) 13 m (D) 14.5m

14. The number of small cubes with edge 10 cm that can be accommodated in a cubical box of edge 1 m is

- (A) 10 (B) 100 (C) 1000 (D) 10000

15. A uniform circular path of width 4 m is laid out around a circular park of radius 48m. Find the area of the circular path.

- (A) 1256 m² (B) 1255 m² (C) 400 m² (D) 1254 m²

16. A flower garden is in the shape of a rhombus. The length of the diagonals are 18 m and 25 m. Find the area of the flower garden.

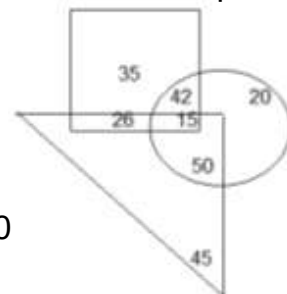
- (A) 450 m² (B) 225 m² (C) 324 m² (D) 18 m²

17. If a square and a rhombus stand on the same base, then the ratio of the area of the square and the rhombus is

- (A) greater than 1 (B) equal to 1
(C) equal to $\frac{1}{2}$ (D) equal to $\frac{1}{4}$

18. The sum of numbers common to two diagrams is

- (A) 118 (B) 110 (C) 108 (D) 130



ANSWER KEYS:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
D	C	C	A	C	B	C	D	B	C	B	D	B	C	A	B	B	A